Aaron M. Bornstein **Updated: August 2, 2026**

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Current position

2019- Assistant Professor, Department of Cognitive Sciences
Faculty, Institute for Mathematical Behavioral Sciences
Fellow, Center for the Neurobiology of Learning and Memory
University of California, Irvine

Previous appointments

2019-2025 Assistant Professor, Department of Cognitive Sciences, UC Irvine
2018 Associate Research Scholar, Princeton Neuroscience Institute
2013-2018 Postdoctoral Research Associate, Princeton Neuroscience Institute
2007-2013 Graduate researcher, New York University
2006 Research assistant, Deicken lab, UCSF/VA
2005-2007 Research assistant, Wagner lab, Stanford

Education

2013 Ph.D., Cognition & Perception, **New York University**
2003 S.B., Mathematics (additional concentration in Economics)
Massachusetts Institute of Technology

Publications

(*∗ = Equal contribution*; *+ = UCI Lab member.*)

Preprints

PP5 Leonard BT+, Martínez-Ortíz MA, Bock J, Zhang Y+, Flores S, Taylor DV, Glynn LM, Davis EP, Stern HS, Baram TZ, Hartley CA, Yassa MA*, **Bornstein AM***. Anhedonia buffers the effects of early-life unpredictability on threat-reward decision-making. *bioRxiv*.
doi:10.64898/2026.05.16.725643

PP4 Wang W, Chierchia G, Cooley BJ, Chang T, **Bornstein AM**, Schweizer S. Autobiographical Memory Specificity Shapes Navigation of Social Uncertainty in Depression. *PsyArXiv*.
doi:10.31234/osf.io/z6vej

PP3 Yoo J+, **Bornstein AM**. Temporal dynamics of model-based control reveal arbitration between multiple task representations. *PsyArXiv*.
doi:10.31234/osf.io/sgcy5

PP2 Zhang Y+*, Chen Y+*, Chen XR+, Harhen NC+, Glynn LA, Davis EP, Baram TZ, Risbrough VB, Stout DA, **Bornstein AM**. A specific computational role for early-life, and not lifelong, stress in decision-making under uncertainty. *bioRxiv*. doi:10.64898/2026.06.01.729407

PP1 Zhou D+, Noh SM+, Harhen NC+, Banavar NV+, Kirwan B, Yassa MA, **Bornstein AM**. A compressed code for memory discrimination. *bioRxiv*. doi:10.1101/2025.10.12.681901

Peer-reviewed journal articles

JP28 Harhen NC+, **Bornstein AM***, Hartley CA* (2026). Developmental changes in memory structure and precision alter the use of retrieved episodes during decisions for reward. *Proceedings of the National Academy of Sciences*, 123(31):e2525494123. doi:10.1073/pnas.2525494123

JP27 Harhen NC+, Budiono R, Hartley CA*, **Bornstein AM*** (2026). Structure inference in complex environments improves from childhood to adulthood. *Developmental Science*, 29(3):e70163. doi:10.1111/desc.70163

JP26 Noh SM+*, Zhou D+*, Cooper KW, Guo S+, Dinh ET+, **Bornstein AM** (2026). Sparsity and memory constraints interact with training sequence to bias learning of associative maps. *Neuropsychologia*, 225:109396. doi:10.1016/j.neuropsychologia.2026.109396

JP25 Noh SM+*, Cooper KW*, Guo S+, Zhou D+, Stark CEL, **Bornstein AM** (2026). Multi-step inference can be improved across the human lifespan with individualized memory interventions. *The Journals of Gerontology, Series B: Psychological Sciences and Social Sciences*, 81(5):gbag038. doi:10.1093/geronb/gbag038

JP24 Hadj-Amar B, **Bornstein AM**, Guindani M, Vannuci M (2025). Discrete Autoregressive Switching Processes with Cumulative Shrinkage Priors for Graphical Modeling of Time Series Data. *Journal of Computational and Graphical Statistics*. doi:10.1080/10618600.2025.2551271

JP23 Schetzsle B, Lee J, **Bornstein AM**, Shahbaba B, Guindani M (2025). A Bayesian Time-Varying Psychophysiological Interaction Model. *Data Science in Science*, 4(1):2519436. doi:10.1080/26941899.2025.2519436

JP22 Khoudary A+, Peters MAK*, **Bornstein AM*** (2025). Reasoning goals and representational decisions in computational cognitive neuroscience: lessons from the drift diffusion model. *European Journal of Neuroscience*, 61:e7009. doi:10.1111/ejn.70098

JP21 Banavar NV+, Noh SM+, Wahlheim CN, Cassidy BS, Kirwan CB, Stark CEL, **Bornstein AM** (2024). A response time model of the three-choice Mnemonic Similarity Task provides stable, mechanistically interpretable individual-difference measures. *Frontiers in Human Neuroscience*, 18:137928. doi:10.3389/fnhum.2024.137928

JP20 Yoo J+, Chrastil ER, **Bornstein AM** (2024). Cognitive graphs: Representational substrates for planning. *Decision*, 11(4), 537-556. doi:10.1037/dec0000249

JP19 Chen J, **Bornstein AM** (2024). The causal structure and computational value of narratives. *Trends in Cognitive Sciences*, 28(8):769-781.
doi:10.1016/j.tics.2024.04.003

JP18 Harhen NC+, **Bornstein AM** (2024). Interval timing as a computational pathway from early life adversity to affective disorders. *Topics in Cognitive Science*, 16(2024):92-112.
doi:10.1111/tops.12701

JP17 Noh SM+, Singla UK, Bennett IJ, **Bornstein AM** (2023). Memory precision and age differentially predict the use of decision-making strategies across the lifespan. *Scientific Reports*, 13:17014.
doi:10.1038/s41598-023-44107-5

JP16 **Bornstein AM**, Aly M, Feng SF, Turk-Browne NB, Norman KA, Cohen JD (2023). Associative memory retrieval modulates upcoming perceptual decisions. *Cognitive, Affective, & Behavioral Neuroscience*, 23:645665.
doi:10.3758/s13415-023-01092-6
doi:10.18112/openneuro.ds001614.v1.0.1

JP15 Harhen NC+, **Bornstein AM** (2023). Overharvesting in human patch foraging reflects rational structure learning and adaptive planning. *Proceedings of the National Academy of Sciences*, 120(13):e2216524120.
doi:10.1073/pnas.2216524120

JP14 Otto AR, Devine S, Schultz E, **Bornstein AM***, Louie K* (2022). Context-dependent choice and evaluation in real-world consumer behavior. *Scientific Reports*, 12:17744.
doi:10.1038/s41598-022-22416-5
doi:10.17605/OSF.IO/EC5DX

JP13 Rmus M, Ritz H, Hunter LE, **Bornstein AM***, Shenhav A* (2022). Humans can navigate complex graph structures acquired during latent learning. *Cognition*, 225:105103.
doi:10.1016/j.cognition.2022.105103

JP12 Wang S, Feng SF, **Bornstein AM** (2021). Mixing memory and desire: How decisions for reward depend on the dynamics and content of memory reinstatement. *Wiley Interdisciplinary Reviews: Cognitive Science*, e1581.
doi:10.1002/wcs.1581
Featured on the cover.
Recognized as one of the top-ten most-cited papers from this journal/year.

JP11 Rouhani N, Norman KA, Niv Y, **Bornstein AM** (2020). Reward prediction errors create event boundaries in memory. *Cognition*, 203:104269.
doi:10.1016/j.cognition.2020.104269

JP10 **Bornstein AM***, Pickard H* (2020). Chasing the first high: Memory sampling in drug choice. *Neuropsychopharmacology*, 45(6):907-915.
doi:10.1038/s41386-019-0594-2 **Featured on the cover.**

JP9 Kane GA, **Bornstein AM**, Shenhav A, Wilson RC, Daw ND, Cohen JD (2019). Rats exhibit similar biases in foraging and intertemporal choice tasks. *eLife*, 8:e48429.
doi:10.7554/eLife.48429

JP8 Millner AJ, den Ouden HEM, Gershman SJ, Glenn CR, Kearns J, **Bornstein AM**, Marx BP, Keane TM, Nock MK (2019). Suicidal thoughts and behaviors are associated with an increased decision-making bias for active responses to escape aversive states. *Journal of Abnormal Psychology*, 128(2):106-118.
doi:10.1037/abn0000395

JP7 Hoskin AN, **Bornstein AM**, Norman KA, Cohen JD (2018). Refresh my memory: Episodic memory reinstatements intrude on working memory maintenance. *Cognitive, Affective, & Behavioral Neuroscience*, 19:338-354.
doi:10.3758/s13415-018-00674-z
doi:10.18112/openneuro.ds001576.v1.0.0

JP6 **Bornstein AM**, Khaw MW, Shohamy D, Daw ND (2017). Reminders of past choices bias decisions for reward in humans. *Nature Communications*, 8:15958.
doi:10.1038/ncomms15958

JP5 **Bornstein AM**, Norman KA (2017). Reinstated episodic context guides sampling-based decisions for reward. *Nature Neuroscience*, 20:997-1003.
doi:10.1038/nn.4573
doi:10.18112/openneuro.ds001607.v1.0.1

JP4 **Bornstein AM**, Daw ND (2013). Cortical and hippocampal correlates of deliberation during model-based decisions for rewards in humans. *PLoS Computational Biology*, 9(12):e1003387.
doi:10.1371/journal.pcbi.1003387

JP3 **Bornstein AM**, Daw ND (2012). Dissociating hippocampal and striatal contributions to sequential prediction learning. *European Journal of Neuroscience*, 35:1011-1023.
doi:10.1111/j.1460-9568.2011.07920.x

JP2 **Bornstein AM**, Daw ND (2011). Multiplicity of control in the basal ganglia: computational roles of striatal subregions. *Current Opinion in Neurobiology*, 21(3):374-380.
doi:10.1016/j.conb.2011.02.009

JP1 Preston AR, **Bornstein AM**, Hutchinson JB, Gaare ME, Glover GH, Wagner AD (2010). High-resolution fMRI of content-sensitive subsequent memory responses in human medial temporal lobe. *Journal of Cognitive Neuroscience*, 22:156-173.
doi:10.1162/jocn.2009.21195

Awards

2024	CNLM Exceptional Mentor Award
2023	Election to the Memory Disorders Research Society
2020	Association for Psychological Science "Rising Star" Award
2020	Brain and Behavior Research Foundation NARSAD Young Investigator Award
2005,6,8	Honorable mention, NSF Graduate Research Fellowship
2000	Runner up, MIT $50k Business Plan competition

Teaching/Service

UC Irvine

Spring 2027	Neural Data Analysis (Undergraduate)
Fall 2026	Cognitive Maps (Graduate; cross-listed with Critical Theory Emphasis)
Fall 2025	Resisting "Neuro-narratives" (Undergraduate)
Winter 2025,2027	LIFTED UC Irvine Prison Education Program (Human Memory)

Spring 2023-2026,Winter 2027 Human Memory (Undergraduate)

Spring 2023-24,Fall 2025 Decision making (Graduate)

Spring 2021-22 Decision making & Problem solving (Graduate; with Prof. Zygmunt Pizlo)

Spring 2020 Topics in Reinforcement Learning (Graduate; with Prof. Mimi Liljeholm)

Spring 2019-22 Research in Exp Psych (Undergraduate; with Prof. Nadia Chernyak)

Winter 2019-22 Advanced Experimental Psychology (Undergraduate)

Other teaching/mentorship

Fall 2020 Neuromatch academy mentor

Summer 2020 Neuromatch academy group leader

Summer 2020 GSMI Cientifico Latino

2019 MEET alumni mentor

Summer 2018 Computational & Cognitive Neuroscience Summer School, Suzhou, China

2016-2018 Princeton Prison Teaching Initiative

Summer 2007,08 MIT Middle East Education through Technology (MEET), Jerusalem

University/Department service

2024- UCI Research Imaging Equipment Committee

2022-2025 Cognitive Sciences Colloquium committee, faculty advisor

2021- Irvine Faculty Association, Executive board member; 2023-2025 Treasurer; 2025- Co-Chair

2020 CNLM "Evening to Remember" organizational committee

2019- First-Generation Faculty Initiative

2019- UCI Prison Education Program, advisory board, curriculum committee

2019,23 Cognitive Sciences faculty search committee

2018,20,22,24 Cognitive Sciences PhD admissions committee

Organizational/Field service

2025- Associate Editor, *Cognitive Science.*

2025- Justice, Diversity, Equity, and Inclusion Committee, *Memory Disorders Research Society.*

2025- Advisor, *All People's Health Collective.*

2024- Editorial board, *Translational Neuroscience.*

2023 UCI Conte Center annual symposium, organized with the Conte Center Team.

2022 NSF/Simons NeuroDataScience workshop, Co-Organizer (with Norbert Fortin and Babak Shahbaba)

2021 Center for Neurobiology of Learning and Memory Spring Meeting Co-organizer (with Lulu Chen)

2018 Princeton Neuroscience Institute "Inside-Out" seminar series Co-organizer (with Ahmed El Hady)

2018 "Goal-Directed Decision Making: Computations and Circuits" *Elsevier* Co-editor (with Richard Morris & Amitai Shenhav)

2015 COSYNE Workshop "Memory in action: The role(s) of the hippocampus in decisions for reward" Co-organizer (with G. Elliott Wimmer)